Unit 3 - Calculus
3.3 Integrals

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Outline

1. Why Integrals?
2. Distance from Velocity
3. Riemann Sums
4. The Definite Integral
5. Integrals from Data
6. Antiderivatives and the Fundamental Theorem
7. Review

Why Integrals?

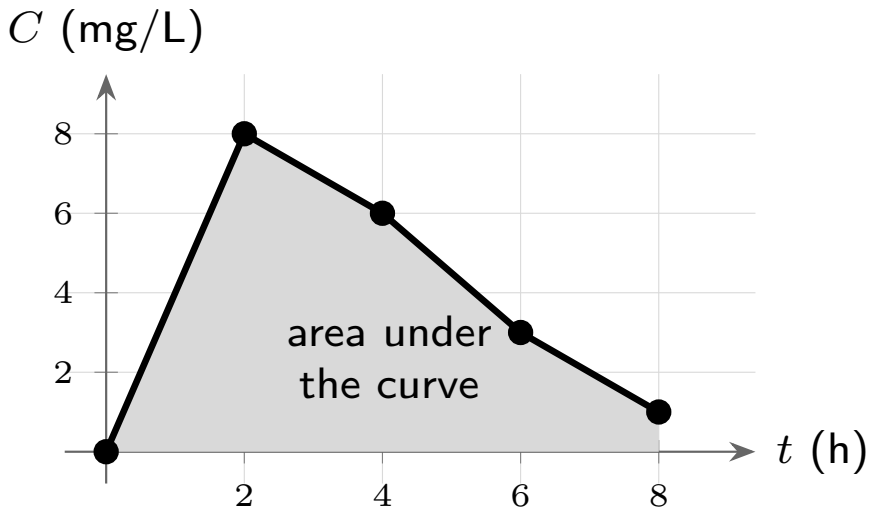
A Problem to Motivate Us

Unit 3-1 went one way

position $x(t)$ $\xrightarrow{\text{derivative}}$ velocity $v(t)$

Today runs it backward

- Only the **speedometer** $v(t)$: how **far** did the car go?
- Only a patient's **blood draws**: how much drug did the body **see**?



Five blood draws after one dose.

The one idea behind both questions

A **rate** times a **time** is an **amount**. When the rate keeps changing, the amount is the **area under the rate's graph**. That area is an **integral**.

Where This Shows Up in Data Science

Integrals underneath the tools

- **Pharmacokinetics.** The **AUC** (area under the concentration-time curve) is the standard measure of total drug exposure. It sets doses, and a generic drug is approved only if its AUC matches the brand's.
- **Probability.** For a continuous measurement (blood pressure, a lab value), the probability of landing in a range is the **area under a density curve**. You will meet this in your statistics courses.
- **Evaluating classifiers.** A model's **ROC AUC** is an area under a curve: about 0.5 for a coin flip, 1 for a perfect model.
- **Totals from rates.** Energy from a power meter (kW $\to$ kWh), fluid from an IV pump's rate (mL/h $\to$ mL), new cases from a daily case rate.

What We Will Build

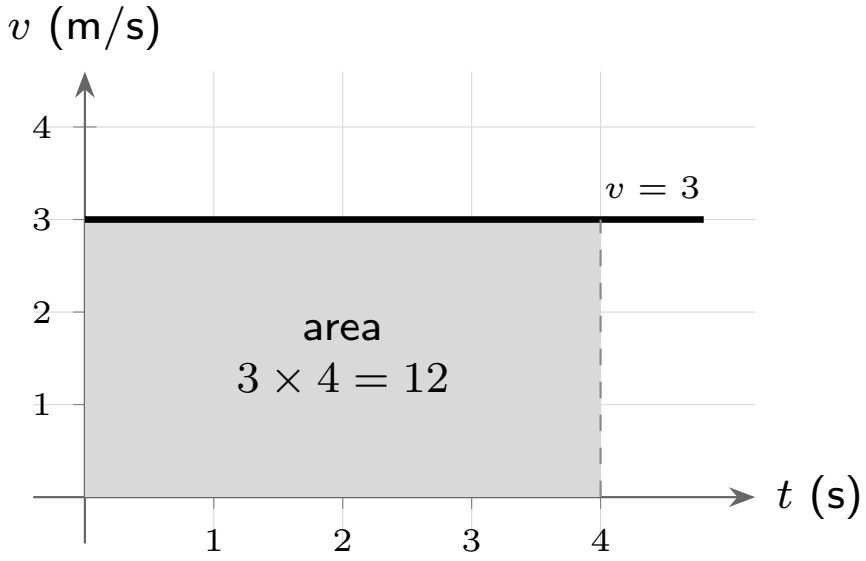
The plan

1. **Distance from velocity**: distance is the area under the velocity graph.
2. **Riemann sums**: approximate any area with rectangles (left, right, midpoint).
3. The **definite integral**: the limit of those sums, its notation, signed area, units.
4. **Integrals from data**: the trapezoid rule, AUC from blood draws, and Python.
5. **Antiderivatives and the Fundamental Theorem**: integration undoes differentiation, which gives exact areas with no rectangles at all.

Notebooks for this lecture: `U3-8_Integrals-1` to `-4`; `U3-8_Integrals-5` takes the same idea to volume under a surface.

Distance from Velocity

Constant Velocity: Distance Is a Rectangle



A car at a steady 3 m/s for 4 s

- Distance = velocity $\times$ time:

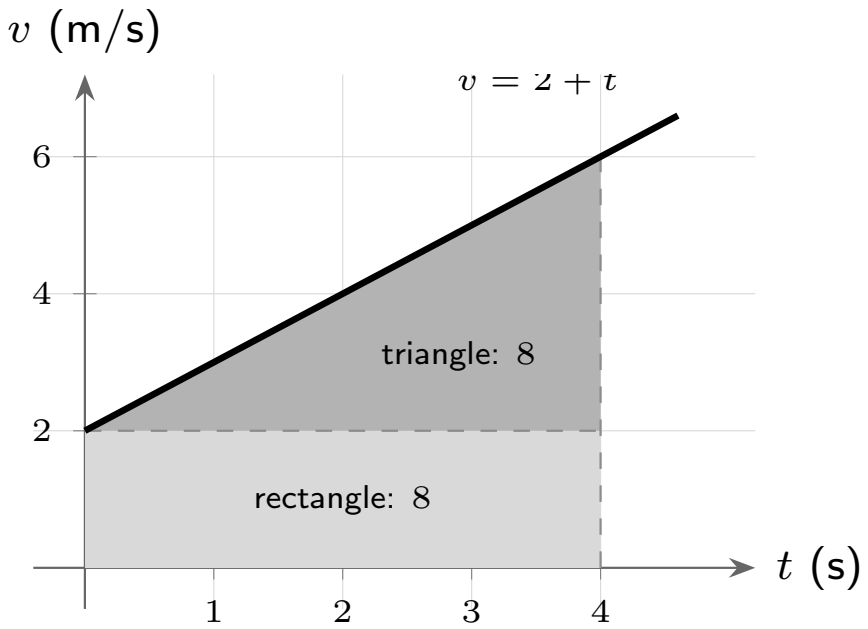
$$d = 3 \cdot 4 = 12 \text{ m}.$$

- On the graph, that product is the **rectangle** under the line: height 3, width 4.
- Units agree: $\frac{\text{m}}{\text{s}} \cdot \text{s} = \text{m}$.

Key idea

Distance traveled = **area under the velocity graph**. Same for an IV pump: 100 mL/h for 3 h is a rectangle of area 300 mL.

Speeding Up Steadily: Rectangle Plus Triangle



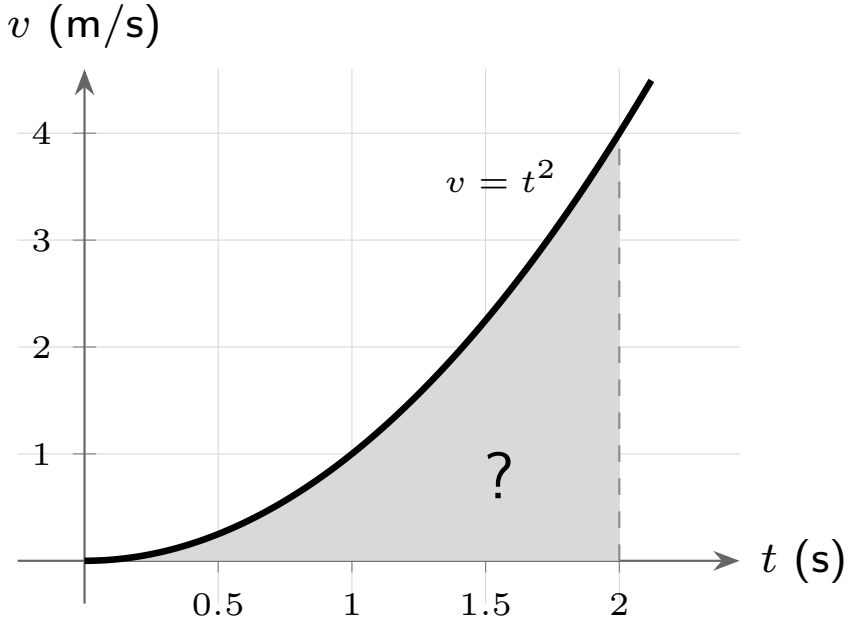
$v(t) = 2 + t$ for $t = 0$ to 4 s

- No single velocity, so no single product.
- But the area is still easy geometry:
  - rectangle: $2 \times 4 = 8$
  - triangle: $\frac{1}{2} \times 4 \times 4 = 8$
- Distance: $8 + 8 = 16$ m.

Check

The velocity climbs evenly from 2 to 6, so its average is 4 m/s, and $4 \times 4 = 16$ m. The area idea still works when velocity changes.

A Curved Velocity: Geometry Runs Out



A car pulling away: $v(t) = t^2$

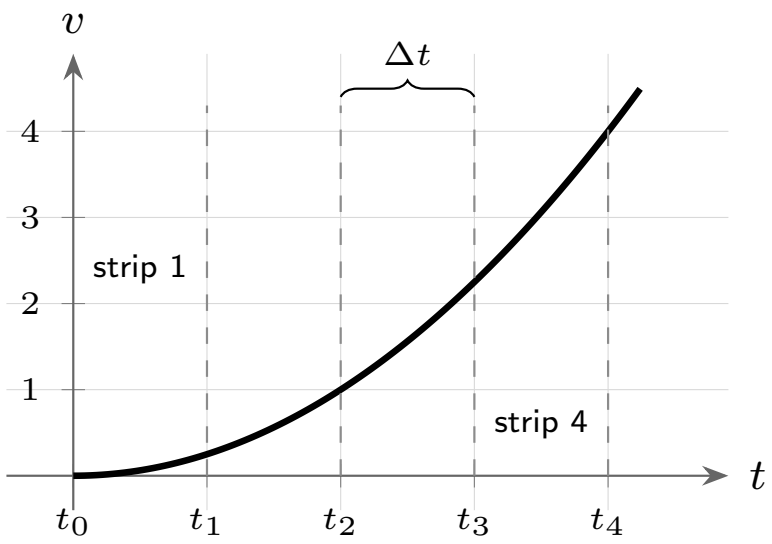
- How far in the first 2 seconds?
- The answer is still the area under the graph.
- But the top edge is **curved**: no rectangle, triangle, or circle formula fits.

The calculus move: solve a simpler problem

- Rectangles we **can** do. Cover the region with thin ones and add them up.
- Make them thinner until the error disappears.
- $v = t^2$ on $[0, 2]$ is the running example for the rest of the lecture.

Riemann Sums

Slice the Interval into Strips



Step 1: cut $[a, b]$ into n strips

- Equal widths: $\Delta t = \frac{b-a}{n}$.
- Cut points: $t_i = a + i\,\Delta t$, from $t_0 = a$ to $t_n = b$.
- Car: $[0, 2]$ with $n = 4$, so $\Delta t = 0.5$; cut points $0, 0.5, 1, 1.5, 2$.

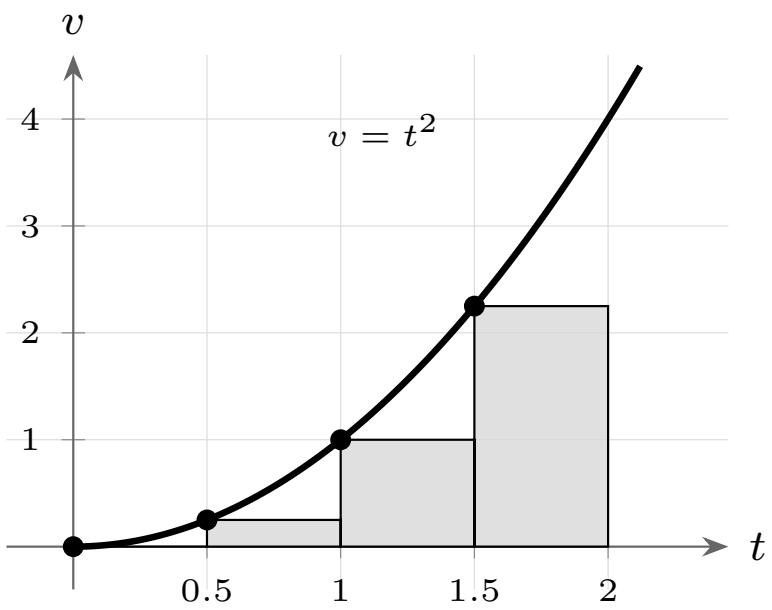
Step 2: one rectangle per strip

Width Δt; height = the function's value at **one chosen point** in the strip: its left end, right end, or middle.

Notebook `U3-8_Integrals-1_RiemannSums`, §1

The Left Endpoint Rule

heights from the **left** ends



Heights at the left ends

strip	**left end**	**height** t^2
1	0	0
2	0.5	0.25
3	1	1
4	1.5	2.25
	sum	3.5

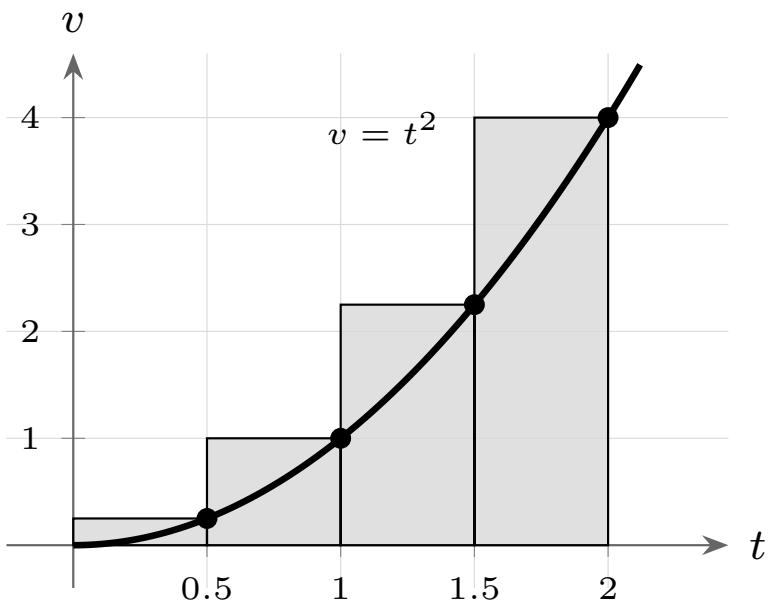
Area $\approx 3.5 \times 0.5 = \mathbf{1.75}$ m.

Watch out: too low

$v = t^2$ is **increasing**, so each strip's left end is its lowest point. Every rectangle sits **under** the curve, and 1.75 is an **underestimate**.

The Right Endpoint Rule

heights from the **right** ends



Heights at the right ends

strip	**right end**	**height** t^2
1	0.5	0.25
2	1	1
3	1.5	2.25
4	2	4
	sum	7.5

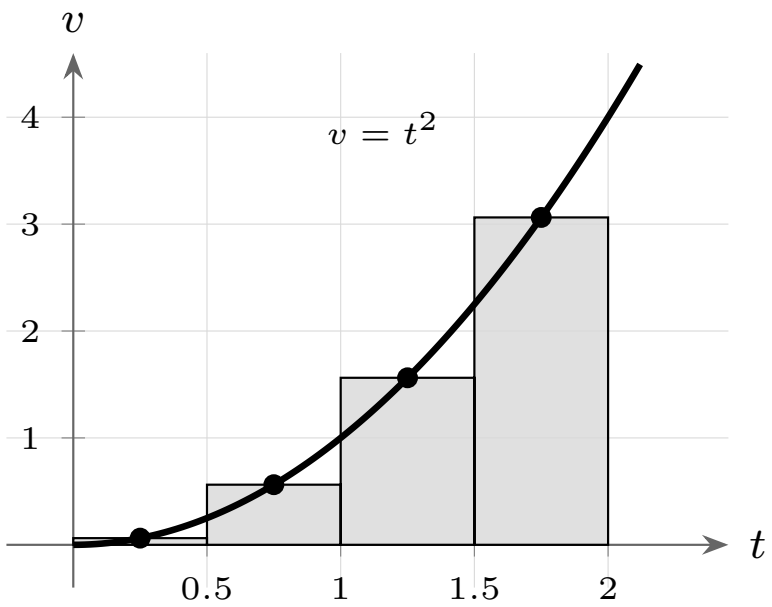
Area $\approx 7.5 \times 0.5 = \mathbf{3.75}$ m.

Watch out: too high

Now each strip uses its **highest** point, so every rectangle pokes **above** the curve: 3.75 is an **overestimate**. The true area is somewhere between 1.75 and 3.75.

The Midpoint Rule

heights from the **middles**



Heights at the middles

strip	**middle**	**height** t^2
1	0.25	0.0625
2	0.75	0.5625
3	1.25	1.5625
4	1.75	3.0625
	sum	5.25

Area $\approx 5.25 \times 0.5 = \mathbf{2.625}$ m.

Why the middle does better

Inside each strip the rectangle is too high on one side and too low on the other. The two errors **nearly cancel**, so the same 4 rectangles land much closer to the truth.

The Riemann Sum

Definition (named for Bernhard Riemann, 19th century)

For f on $[a, b]$ cut into n strips of width $\Delta x = \frac{b-a}{n}$, pick a point x_i^* in strip i. The **Riemann sum** is

$$\sum_{i=1}^{n} f(x_i^*)\,\Delta x \;=\; f(x_1^*)\,\Delta x + f(x_2^*)\,\Delta x + \cdots + f(x_n^*)\,\Delta x.$$

Reading it

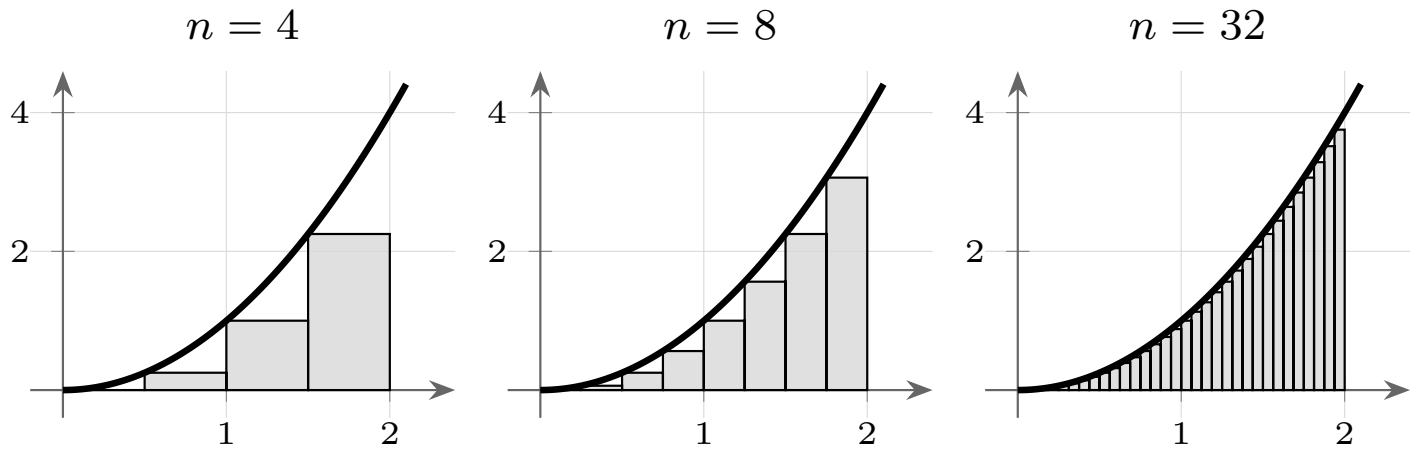
- $\sum_{i=1}^{n}$: add up the strips (Unit 1-4).
- $f(x_i^*)$: **height** of rectangle i.
- Δx: its **width**.
- Strip i runs from x_{i-1} to x_i.

The three rules

rule	**chosen point** x_i^*
left	x_{i-1}
right	x_i
midpoint	$\frac{1}{2}(x_{i-1} + x_i)$

Notebook `U3-8_Integrals-1_RiemannSums`, §2

More Rectangles, Closer Answers



Left rule for $v = t^2$ on $[0, 2]$: the gaps above the rectangles shrink as n grows.

Area under $v = t^2$ on $[0, 2]$

n	**left**	**right**	**midpoint**
4	1.7500	3.7500	2.6250
8	2.1875	3.1875	2.6563
16	2.4219	2.9219	2.6641
100	2.6268	2.7068	2.6666

What the table says

- All three close in on **one number**, about 2.667.
- Left and right squeeze it from below and above.
- Midpoint gets there fastest.

Notebook `U3-8_Integrals-1_RiemannSums`, §3 and §4

Exercise: Riemann Sums

Try it

A patient's IV is being tapered off: the drip rate is $f(x) = 4 - x^2$ mL/min for $x = 0$ to 2 minutes.

1. With $n = 4$ strips, find Δx and the cut points.
2. Compute the **left** and **right** Riemann sums.
3. Which one is an overestimate, and why? (Sketch the graph.)
4. Compute the **midpoint** sum. What does the number mean for the patient?

Exercise: Solution

Answers

1. $\Delta x = \frac{2-0}{4} = 0.5$; cut points $0, 0.5, 1, 1.5, 2$.
2. Heights $f(0) = 4$, $f(0.5) = 3.75$, $f(1) = 3$, $f(1.5) = 1.75$, $f(2) = 0$.
   - Left: $(4 + 3.75 + 3 + 1.75) \times 0.5 = 12.5 \times 0.5 = 6.25$.
   - Right: $(3.75 + 3 + 1.75 + 0) \times 0.5 = 8.5 \times 0.5 = 4.25$.
3. This time the **left** sum is too high. The rate is **decreasing**, so each strip's left end is its highest point. (The reverse of $v = t^2$.)
4. Middles $0.25, 0.75, 1.25, 1.75$ give heights $3.9375, 3.4375, 2.4375, 0.9375$, which sum to 10.75; times 0.5 gives 5.375.

Meaning: the patient receives about 5.4 mL in those 2 minutes: rate (mL/min) times time (min) is a volume (mL). The exact answer comes at the end of the lecture.

The Definite Integral

From Sums to the Integral

Let the rectangles get infinitely thin

Take more and more strips, $n \to \infty$, so the width $\Delta x \to 0$. The Riemann sums settle on one number: the **definite integral** of f from a to b,

$$\int_a^b f(x)\,dx \;=\; \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\,\Delta x.$$

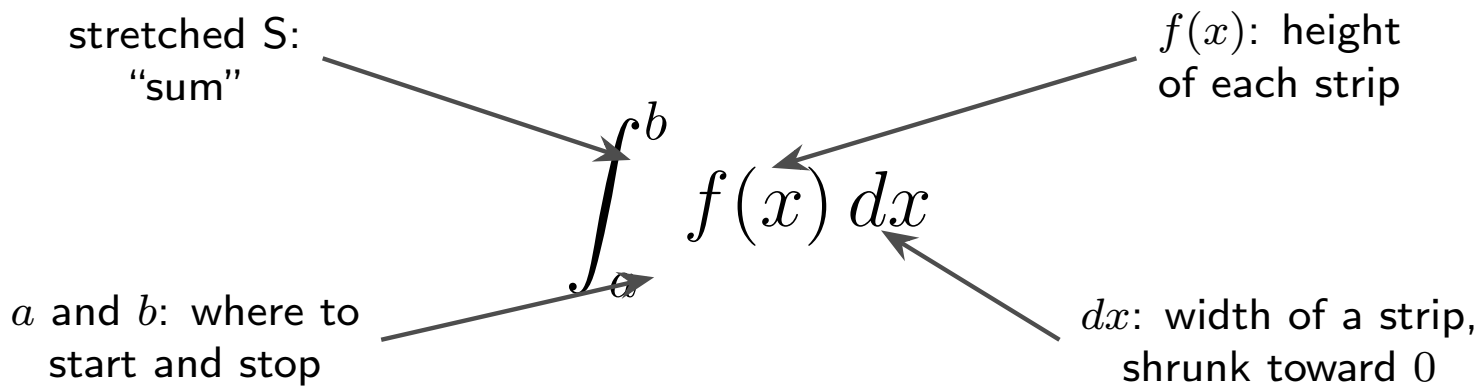
Same move as Unit 3-1

- Derivative: slopes of secants, let $h \to 0$.
- Integral: sums of rectangles, let $\Delta x \to 0$.
- Both: approximate, then let the error shrink to nothing.

Good to know

- For a smooth curve, left, right, and midpoint all reach the **same** limit.
- The integral is a **number**, not a function.
- Car: $\int_0^2 t^2\,dt \approx 2.667$ m.

Reading the Notation

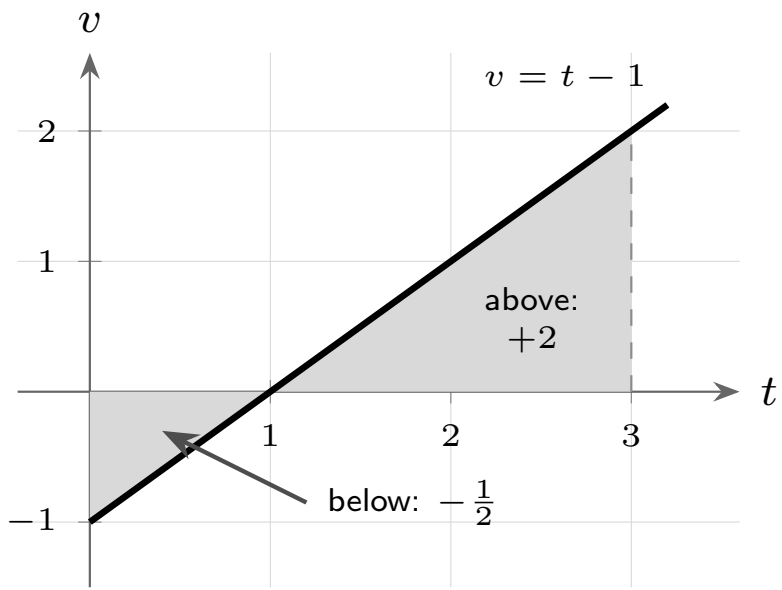


Sum and integral, side by side

	Riemann sum	**integral**
add up	$\sum$	$\int$
height	$f(x_i^*)$	$f(x)$
width	Δx	dx
range	$i = 1$ to n	$x = a$ to b

a and b are the **limits of integration**. The letter after d names the variable: $\int_0^2 t^2\,dt$ adds up along t.

Signed Area: Below the Axis Counts Negative



The rule

- Where $f(x) < 0$, the heights $f(x_i^*)$ are negative, so those rectangles **subtract**.
- $\int_0^3 (t-1)\,dt = 2 - \frac{1}{2} = \frac{3}{2}$.

What it means: net, not total

- $v < 0$ means the car is **backing up**. For $t = 0$ to 1 it backs up $\frac{1}{2}$ m, then drives forward 2 m.
- The integral gives where it **ends up**: $\frac{3}{2}$ m ahead (net displacement).
- Total ground covered is $\frac{1}{2} + 2 = \frac{5}{2}$ m. That needs the areas added **without** signs.

Units: What an Integral Measures

Units of $\int_a^b f(x)\,dx$ = (units of f) $\times$ (units of x)

Each rectangle is a height times a width, so the total carries both units.

Rates in, totals out

f **(height)**	x **(width)**	**integral**	**meaning**
velocity, m/s	time, s	m	distance
IV rate, mL/h	time, h	mL	volume given
drug level, mg/L	time, h	mg·h/L	AUC, exposure
power, kW	time, h	kWh	energy used
new cases per day	time, days	cases	total cases

Rule of thumb

If f is a **rate** (something per unit of x), the integral is the **amount** that piled up.

Exercise: Integrals from Geometry

Try it

Sketch each graph and find the integral with area formulas (no rectangles needed).

1. $\int_0^5 2\,dx$
2. $\int_0^4 x\,dx$
3. $\int_{-2}^2 x\,dx$
4. $\int_0^2 (x-2)\,dx$
5. An IV pump runs at $f(x)$ mL/h, with x in hours. What are the units of $\int_0^6 f(x)\,dx$, and what does it measure?

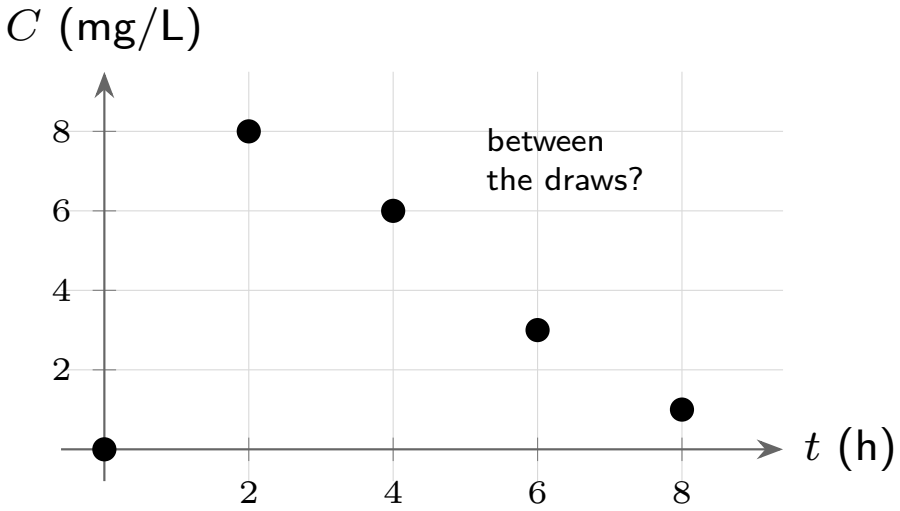
Exercise: Solution

Answers

1. A rectangle, height 2, width 5: area $= 10$.
2. A triangle from $(0, 0)$ to $(4, 4)$: $\frac{1}{2} \times 4 \times 4 = 8$.
3. Two triangles of area $\frac{1}{2} \times 2 \times 2 = 2$: one below the axis (from -2 to 0), one above (from 0 to 2). Signed: $-2 + 2 = 0$. The area is not zero, but the **net** is.
4. The line runs from -2 up to 0, entirely **below** the axis: a triangle of area $\frac{1}{2} \times 2 \times 2 = 2$, counted negative. Integral $= -2$.
5. (mL/h) $\times$ (h) $=$ **mL**: the total volume the patient received in the first 6 hours.

Integrals from Data

When All You Have Is Measurements



Blood draws after one oral dose

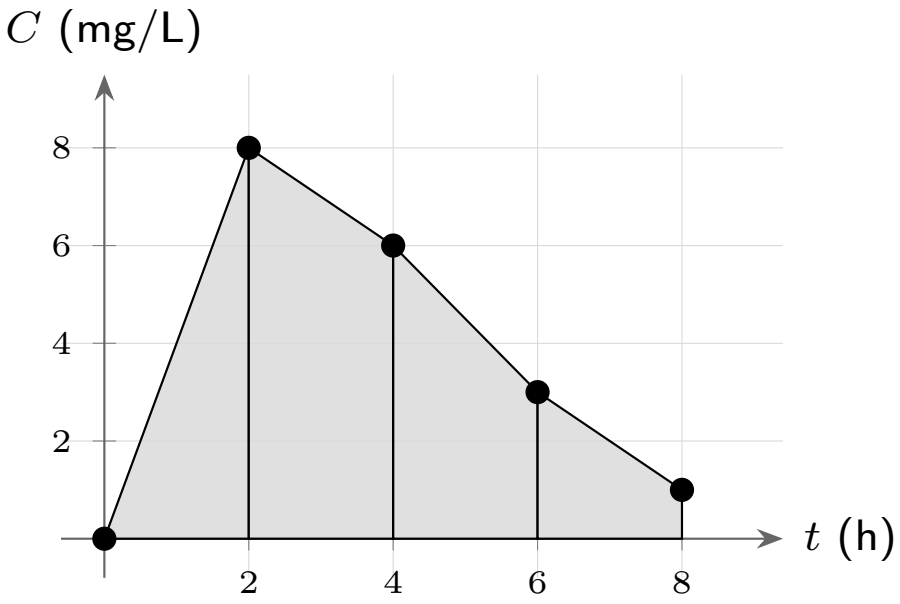
t (h)	0	2	4	6	8
C (mg/L)	0	8	6	3	1

What changes with data

- No formula: we know C only at the draw times.
- Δt is **fixed by the schedule** $(2$ h$)$: the strips cannot get thinner.
- No midpoint rule: nobody measured at $t = 1, 3, 5, 7$.
- Left and right still work: $(0 + 8 + 6 + 3) \times 2 = 34$, $(8 + 6 + 3 + 1) \times 2 = 36$.

Notebook `U3-8_Integrals-3_DataPoints`, §1 and §2

The Trapezoid Rule



Join neighbors with straight lines

- Each strip becomes a **trapezoid**: width Δt, average height $\frac{y_i + y_{i+1}}{2}$.
- Strip area: $\dfrac{y_i + y_{i+1}}{2}\,\Delta t$.
- It is exactly the **average** of the left and right rules.

The trapezoid rule

$$T = \Delta t \left(\tfrac{1}{2}y_0 + y_1 + y_2 + \cdots + y_{n-1} + \tfrac{1}{2}y_n\right)$$

- Uneven draw times (common in real studies) are fine: give each strip its own width.

Worked: AUC from Blood Draws

Strip by strip, $\Delta t = 2$ h

strip	average height	area
0 to 2	$\frac{0+8}{2} = 4$	$4 \times 2 = 8$
2 to 4	$\frac{8+6}{2} = 7$	$7 \times 2 = 14$
4 to 6	$\frac{6+3}{2} = 4.5$	$4.5 \times 2 = 9$
6 to 8	$\frac{3+1}{2} = 2$	$2 \times 2 = 4$
	total	**35**

Check

- Formula:
  $2\left(\frac{1}{2}\cdot 0 + 8 + 6 + 3 + \frac{1}{2}\cdot 1\right)$
  $= 2 \times 17.5 = 35.$
- Average of left and right:
  $\frac{34+36}{2} = 35.$

Answer

- $\text{AUC}_{0 \text{ to } 8\text{h}} \approx 35$ mg·h/L: the patient's total exposure to the drug over the first 8 hours.
- Real studies also estimate the tail after the last draw, since the drug is not fully gone at 8 h.

Notebook `U3-8_Integrals-3_DataPoints`, §4

Integrals from Data in Python

NumPy's `trapezoid`

```
import numpy as np
t = np.array([0, 2, 4, 6, 8])  # hours
C = np.array([0, 8, 6, 3, 1])  # mg/L

np.trapezoid(C, t)    # trapezoid rule
# 35.0
dt = 2                # left and right rules
C[:-1].sum() * dt, C[1:].sum() * dt
# (34, 36)

# a formula: sample it finely
x = np.linspace(0, 2, 101)   # 100 strips
np.trapezoid(x**2, x)
# 2.6668000000000003
```

Reading the call

- `np.trapezoid(y, x)`: **heights first**, then positions (uneven spacing is fine).
- Formula: `linspace` picks the strips. Data: the schedule does.

Watch out

- `...0003` is floating point (Unit 1-5).
- NumPy before 2.0: `np.trapz`.

Notebook `U3-8_Integrals-4_RealData`, §2, a real power meter

Exercise: Brand vs. Generic

Try it

Two formulations of the same drug, blood drawn every hour (mg/L):

t (h)	0	1	2	3	4
brand	0	4	5	3	2
generic	0	6	4	2	1

1. Use the trapezoid rule to find each formulation's AUC from 0 to 4 h.
2. Compute the ratio $\text{AUC}_{\text{generic}}/\text{AUC}_{\text{brand}}$ as a percent.
3. A generic is judged equivalent when this ratio falls within 80% to 125%. Does it pass?
4. The generic reaches a **higher peak**. Does that mean the body sees more drug?

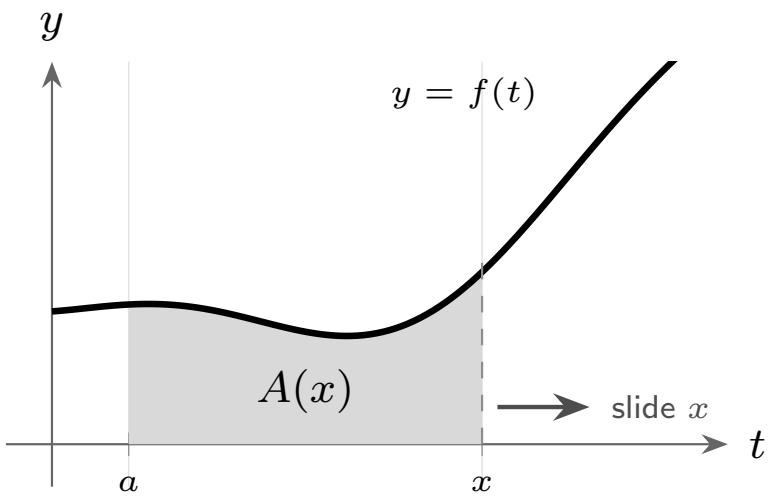
Exercise: Solution

Answers

1. With $\Delta t = 1$, each strip's area is just the average of its two ends:
   - Brand: $\frac{0+4}{2} + \frac{4+5}{2} + \frac{5+3}{2} + \frac{3+2}{2} = 2 + 4.5 + 4 + 2.5 = 13$ mg·h/L.
   - Generic: $\frac{0+6}{2} + \frac{6+4}{2} + \frac{4+2}{2} + \frac{2+1}{2} = 3 + 5 + 3 + 1.5 = 12.5$ mg·h/L.
2. $12.5/13 \approx 0.96$, so about 96%.
3. Yes: 96% sits inside 80% to 125%. (A real approval runs this on many volunteers with a statistical test, and compares peaks too.)
4. No. The generic peaks higher (6 vs. 5) and sooner, but its **total** exposure is slightly **lower**. The peak and the AUC answer different questions: how high, versus how much.

Antiderivatives and the Fundamental Theorem

A Function In, a Function Out



Two kinds of process

- $\int_a^b f(x)\,dx$: function and interval in, **number** out.
- The derivative: function in, **function** out.
- Can integration give a function too?

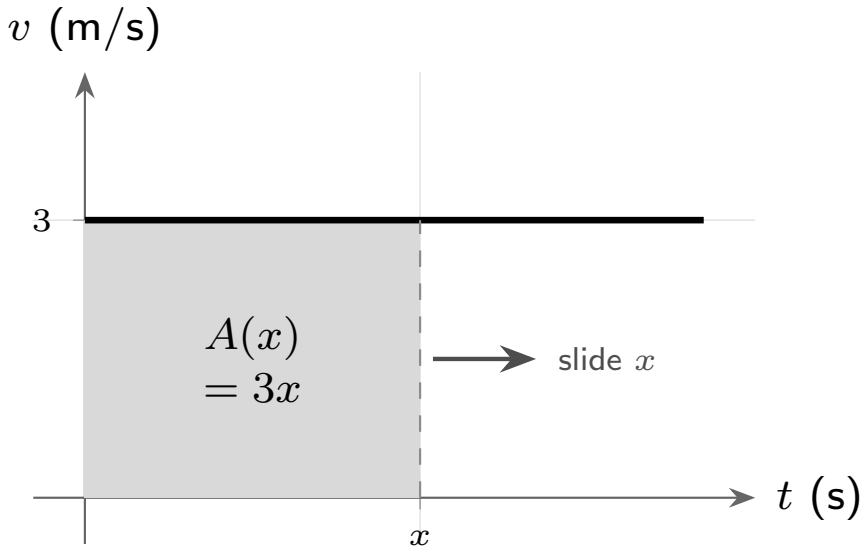
Let the right end move: the "area so far"

$$A(x) = \int_a^x f(t)\,dt$$

- The area from a up to a **movable** x: each x gives a number, so A is a function.
- t inside is a placeholder, since x now names the moving endpoint.

Notebook `U3-8_Integrals-2_FundamentalTheorem`, §1

Constant Velocity: The Distance So Far



$v(t) = 3$ m/s, starting at $a = 0$

- Area so far: a rectangle, height 3, width x.

$$A(x) = 3x$$

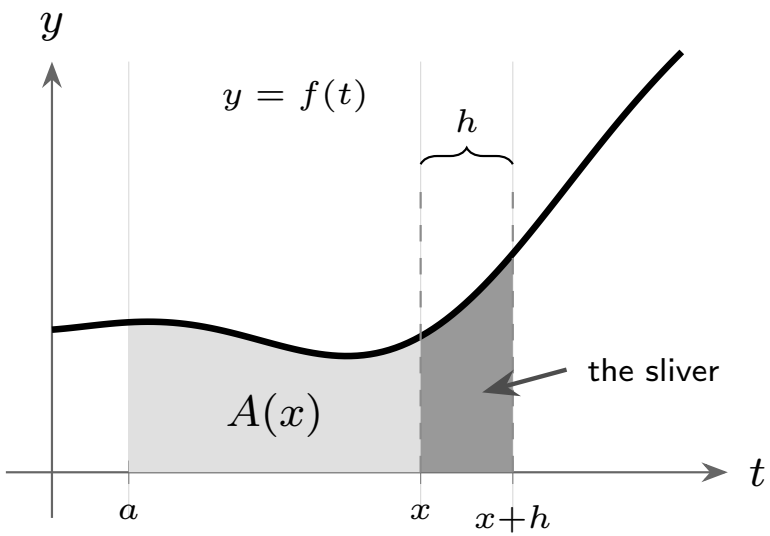
- Differentiate it:

$$A'(x) = 3 = v(x).$$

What just happened

- Differentiating the **area so far** gave back the **function we started with**.
- Of course: distance so far grows at the rate of the velocity. That is what velocity means.

Why $A'(x) = f(x)$: The Thin Sliver



Nudge x by a little h

- The area grows by one thin **sliver**, $A(x+h) - A(x)$.
- The sliver is almost a rectangle: height $f(x)$, width h.

$$A(x+h) - A(x) \approx f(x)\,h$$

- Thinner slivers look more like rectangles.

Divide by h, then let h shrink (Unit 3-1)

$$\frac{A(x+h) - A(x)}{h} \approx f(x) \quad \xrightarrow{h \to 0} \quad A'(x) = f(x)$$

Notebook `U3-8_Integrals-2_FundamentalTheorem`, §2 and §3

Antiderivatives

Definition

F is an **antiderivative** of f if $F'(x) = f(x)$. It runs the derivative **backward**: "what did I differentiate to get f?"

Example: $f(x) = 2x$

- $F(x) = x^2$ works: $(x^2)' = 2x$.
- So do $x^2 + 1$ and $x^2 - 7$: a constant differentiates to 0.
- Every antiderivative is $x^2 + C$ for some constant C.

Notation

- The **indefinite integral** (no limits): $\int 2x\,dx = x^2 + C$.
- C is the **constant of integration**.

Always check

Differentiate your answer: if you get f back, it is right.

A Table of Antiderivatives

Each row reads backward from a derivative rule in Unit 3-1

$f(x)$	$\int f(x)\,dx$	**check: differentiate**
k (a constant)	$kx + C$	$(kx)' = k$
$x^n, \quad n \neq -1$	$\frac{x^{n+1}}{n+1} + C$	$\frac{(n+1)x^n}{n+1} = x^n$
$\frac{1}{x}$	$\ln\lvert x\rvert + C$	$(\ln x)' = \frac{1}{x}$
e^x	$e^x + C$	$(e^x)' = e^x$
$\cos x$	$\sin x + C$	$(\sin x)' = \cos x$
$\sin x$	$-\cos x + C$	$(-\cos x)' = \sin x$

Rules

- Constants factor out: $\int kf = k\int f$.
- Sums split: $\int (f+g) = \int f + \int g$.

Watch out

$n = -1$ would divide by 0 in the power rule. That is why $\frac{1}{x} = x^{-1}$ gets its own row.

The Fundamental Theorem of Calculus

The theorem

1. The area so far is an antiderivative: $\frac{d}{dx}\int_a^x f(t)\,dt = f(x)$.
2. If F is **any** antiderivative of f, then

$$\int_a^b f(x)\,dx = F(b) - F(a), \quad \text{written} \quad \Big[F(x)\Big]_a^b.$$

Why part 2 follows from part 1

- A and F are both antiderivatives, so $A(x) = F(x) + C$.
- No area yet at the start: $A(a) = 0$, so $C = -F(a)$.
- At the end: $A(b) = F(b) - F(a)$.

What it means

- Integration and differentiation **undo each other**.
- Exact areas, no rectangles.
- The $+C$ cancels in $F(b) - F(a)$, so use $C = 0$.

Notebook `U3-8_Integrals-2_FundamentalTheorem`, §4

Back to the Car: Exactly $\frac{8}{3}$

$v = t^2$ on $[0, 2]$

- Antiderivative: $F(t) = \frac{t^3}{3}$, since $\left(\frac{t^3}{3}\right)' = t^2$.
- Fundamental Theorem:

$$\int_0^2 t^2\,dt = \left[\frac{t^3}{3}\right]_0^2 = \frac{8}{3} - 0 = \frac{8}{3}$$

- $\frac{8}{3} \approx 2.667$ m.

Where the sums were heading

n	**midpoint sum**
4	2.6250
8	2.6563
16	2.6641
100	2.6666
exact	$\frac{8}{3} = 2.6667$

And the signed-area example: the signs take care of themselves

$\int_0^3 (t-1)\,dt = \left[\frac{t^2}{2} - t\right]_0^3 = \left(\frac{9}{2} - 3\right) - 0 = \frac{3}{2}$, matching $2 - \frac{1}{2}$ from the picture.

Exercise: Using the Fundamental Theorem

Try it

Find each integral exactly. Check your antiderivative by differentiating.

1. $\displaystyle\int_1^3 2x\,dx$. Then check the answer with geometry.
2. $\displaystyle\int_0^2 (4 - x^2)\,dx$: the tapering IV from the Riemann sum exercise. Which of your sums (6.25, 4.25, 5.375) came closest?
3. A cyclist's velocity is $v(t) = 6t - 3t^2$ m/s for $t = 0$ to 2 s. How far does the cyclist travel? ($v \geq 0$ on this interval.)
4. $\displaystyle\int_1^e \frac{1}{x}\,dx$

Exercise: Solution

Answers

1. $F(x) = x^2$: $\left[x^2\right]_1^3 = 9 - 1 = 8$. Geometry: a trapezoid with heights 2 and 6, width 2: $\frac{2+6}{2} \times 2 = 8$. ✓
2. $F(x) = 4x - \frac{x^3}{3}$: $\left(8 - \frac{8}{3}\right) - 0 = \frac{16}{3} \approx 5.33$ mL. The true value sits between the right (4.25) and left (6.25) sums, and the **midpoint** (5.375) missed by only about 0.04.
3. $F(t) = 3t^2 - t^3$ (check: $F'(t) = 6t - 3t^2$): $F(2) - F(0) = (12 - 8) - 0 = 4$ m. Since $v \geq 0$, net and total distance agree.
4. $F(x) = \ln|x|$: $\ln e - \ln 1 = 1 - 0 = 1$. The area under $\frac{1}{x}$ from 1 to e is exactly 1, one way to define the number e.

Review: Integrals

What we covered: Riemann sums and the definite integral

1. A rate times a time is an amount. When the rate changes, the amount is the **area under the rate's graph**: distance from velocity, exposure from concentration.
2. A **Riemann sum** cuts $[a, b]$ into n strips of width $\Delta x = \frac{b-a}{n}$ and adds up rectangles: $\sum_{i=1}^{n} f(x_i^*)\,\Delta x$. Heights from the left end, right end, or middle. For an increasing f, left is too low and right too high; midpoint is usually best.
3. The **definite integral** is the limit as $n \to \infty$: $\int_a^b f(x)\,dx$. $\int$ is the sum, $f(x)$ the height, dx the width.
4. Area below the axis counts **negative**, so the integral is a **net** amount. Its units are (units of f) $\times$ (units of x).

Review: Integrals

What we covered: data and the Fundamental Theorem

5. With **data**, the sampling schedule fixes Δx. The **trapezoid rule**, the average of left and right, is the standard choice; it is how AUC is computed. In code, `np.trapezoid(y, x)`.
6. The **area so far**, $A(x) = \int_a^x f(t)\,dt$, satisfies $A'(x) = f(x)$: the thin-sliver argument.
7. An **antiderivative** F has $F' = f$, and comes with a $+C$. Check one by differentiating.
8. **Fundamental Theorem:** $\int_a^b f(x)\,dx = F(b) - F(a)$. Integration and differentiation undo each other, and exact areas need no rectangles.

Notebooks: `U3-8_Integrals-1` to `-5`. **Previous:** Unit 3-2, Optimization and Linear Regression. This closes Unit 3, **Calculus**.